Solar Storm Risk Prediction

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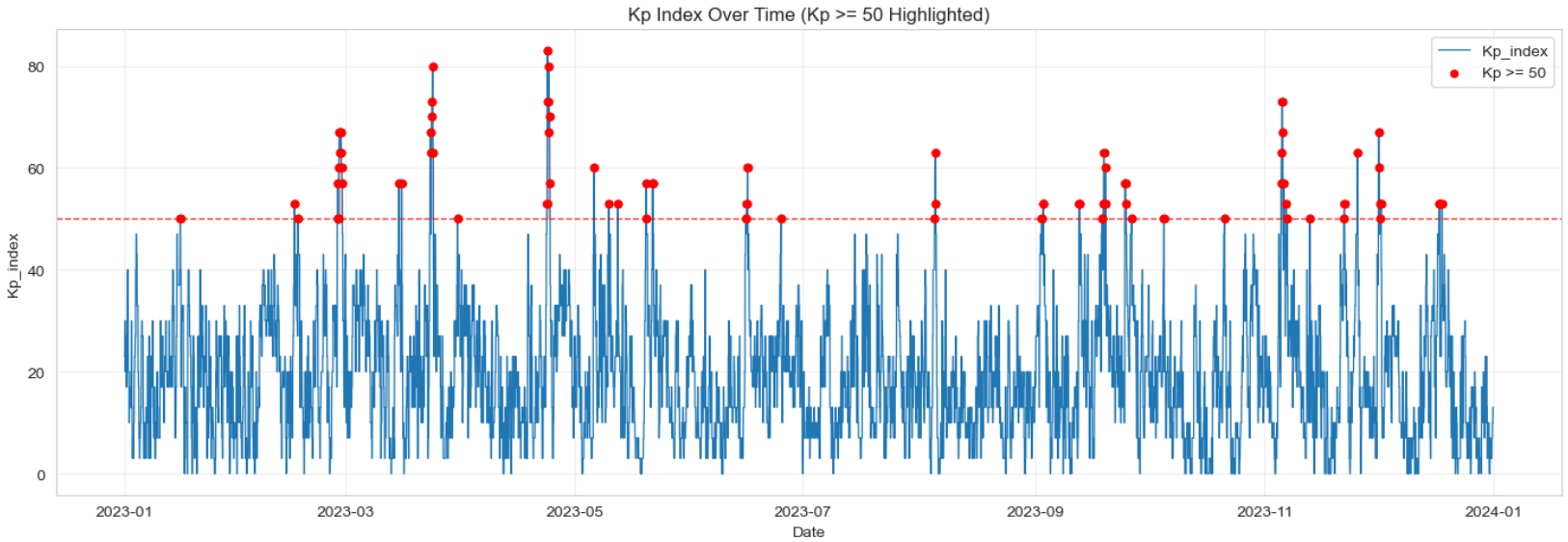
Summary

- Sun emits solar wind towards the Earth
- Can create geomagnetic storms affecting power grids, satellites, and communication systems.
- Useful for operators to receive "early warning" about potential solar storms.
- This project: predict solar storms from incoming solar wind data (~45 minute lead time)

Problem setup

- Solar storms do not happen all the time → rare event prediction
- Weather is simply not 100% predictable.
- False positive (false alarm) is more desirable than false negative → prefer to maximize recall subject to fixed precision.
- Weather prediction goals: recall > .60, precision > .30



Data sources

- NASA and NOAA provide data from satellites situated between the Earth and the Sun.
- NASA: historical time-aligned data
- NOAA: real time solar wind data
- Satellite data can have gaps, so data cleaned

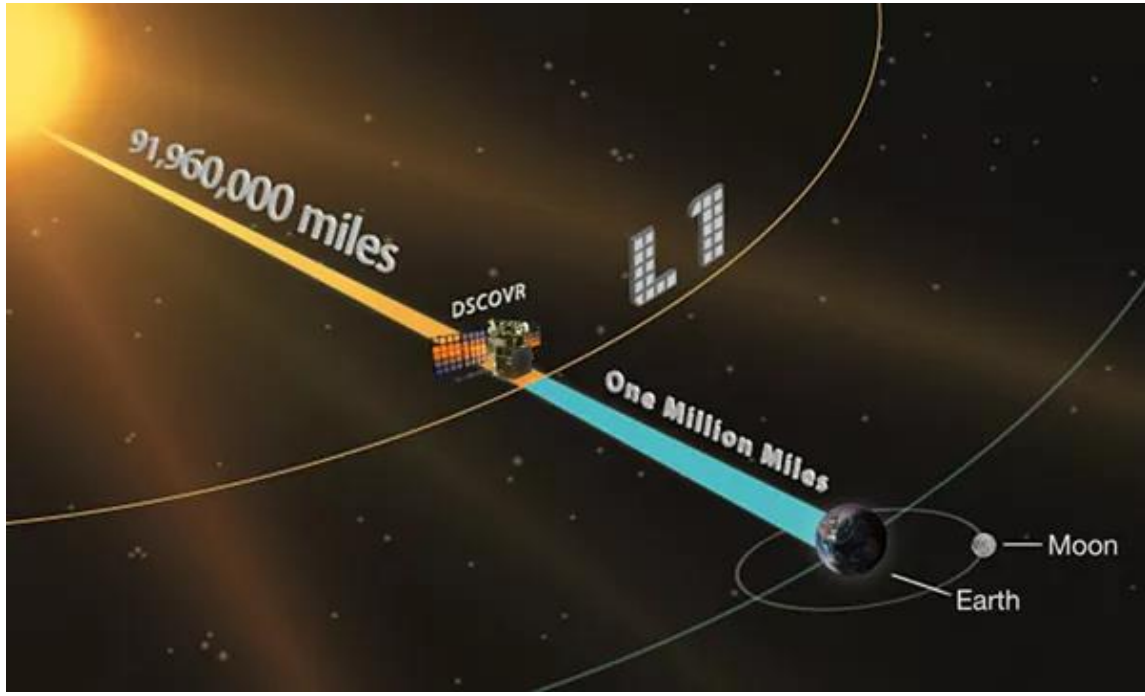


Photo credit: NASA

Feature selection

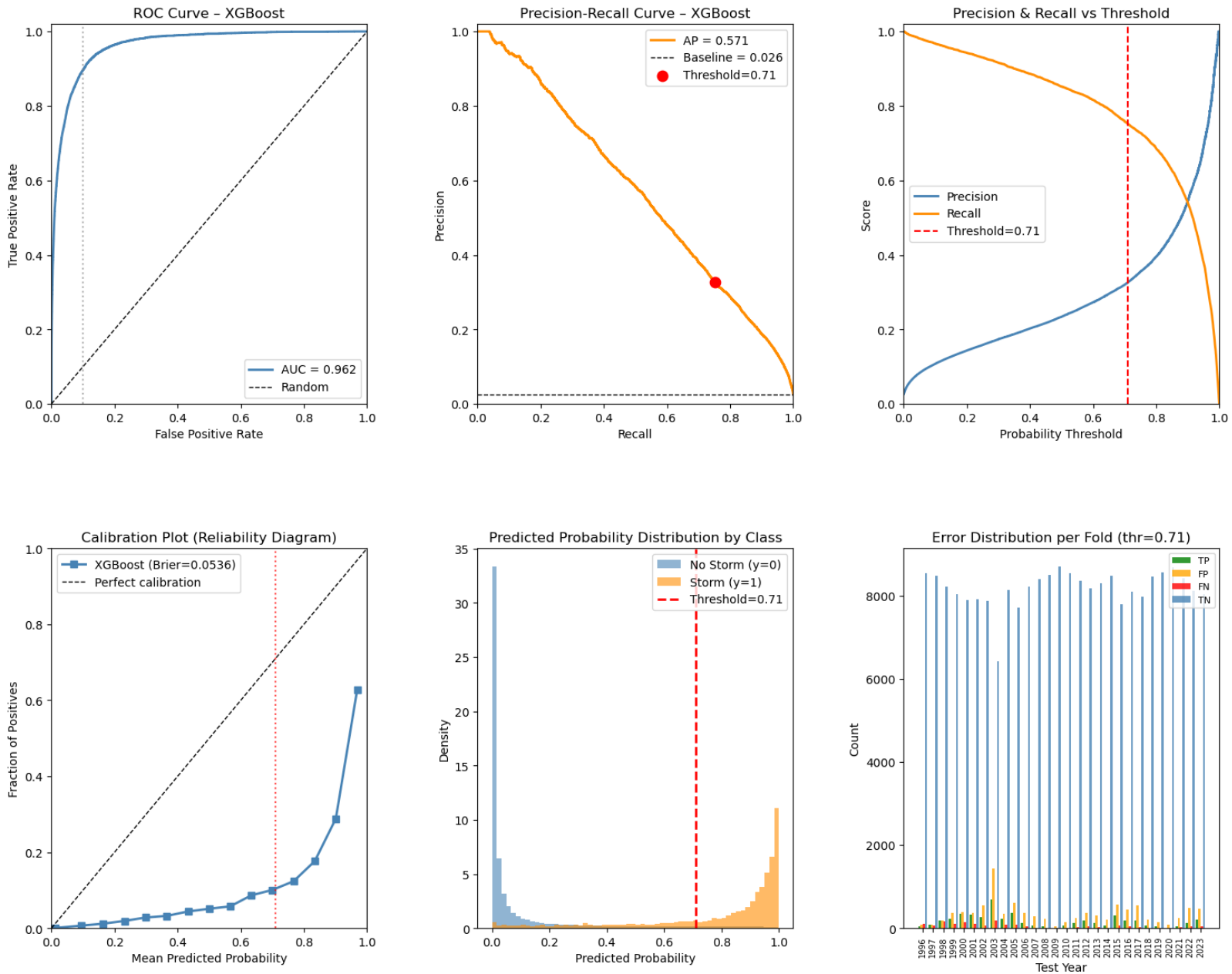
- Physics informed features:
  - Angle between solar wind and Earth's magnetic field
  - Energy flux
  - Rolling averages of measurements
  - Variation in magnetic orientation of solar wind
  - "Newell coupling"
- Some features were highly correlated, so they were dropped to improve model performance
  - Many engineered rolling averages were too highly correlated with raw features.
  - Features encoding solar wind vector had some redundancy and so a few were dropped.

Model Selection

- Time series data
  - Used 20 years of historical data
  - Cross-validation with 1 year folds
- Weather prediction domain
  - Maximize recall with precision >= .30
  - Maintain realistic expectations for results
  - Adjust probability threshold on models
- Hyperparameter tuning using Optuna
- Baselines
  - High precision 0 recall: Dummy classifier always says "No Storm"
  - Other standard models without feature engineering: < 0.3 recall and < 0.6 precision

Results (XGBClassifier)

- Overall Precision: 0.3260
- Overall Recall : 0.7517



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Limitations and potential

- Will give many false positives
- Further time horizon predictions not realistic without additional data sources
  - Optical data from sun
- Model could be deployed with "online" data from NOAA
  - Data ingestion built, but online deployment not developed for lack of time
- Business use case
  - Provide advanced warning of potential geomagnetic storm to grid operators